Removing Camera Fog caused by Water-proof Pouch using Temporal Fusion Techniques

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Agenda

01 Preliminaries

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Preliminaries

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Motivation

Wonderful river tracing, but...



Fog on water-proof pouch







Problem formulation

Given a “foggy moving image”, integrate clue from different frames and restore to “fogless moving image”

Note that our goal is to “restore a foggy image using property of video”, not to “restore a foggy video”, we regard input foggy video as “moving image”

We assume:

- motion of all pixels are the same, that is, motion of the camera.
- motion of the camera will not move back
- motion of the camera is slow enough and at steady speed
- the solid body (e.g. rock) are mainly in the middle of image

Outcome

Before



After



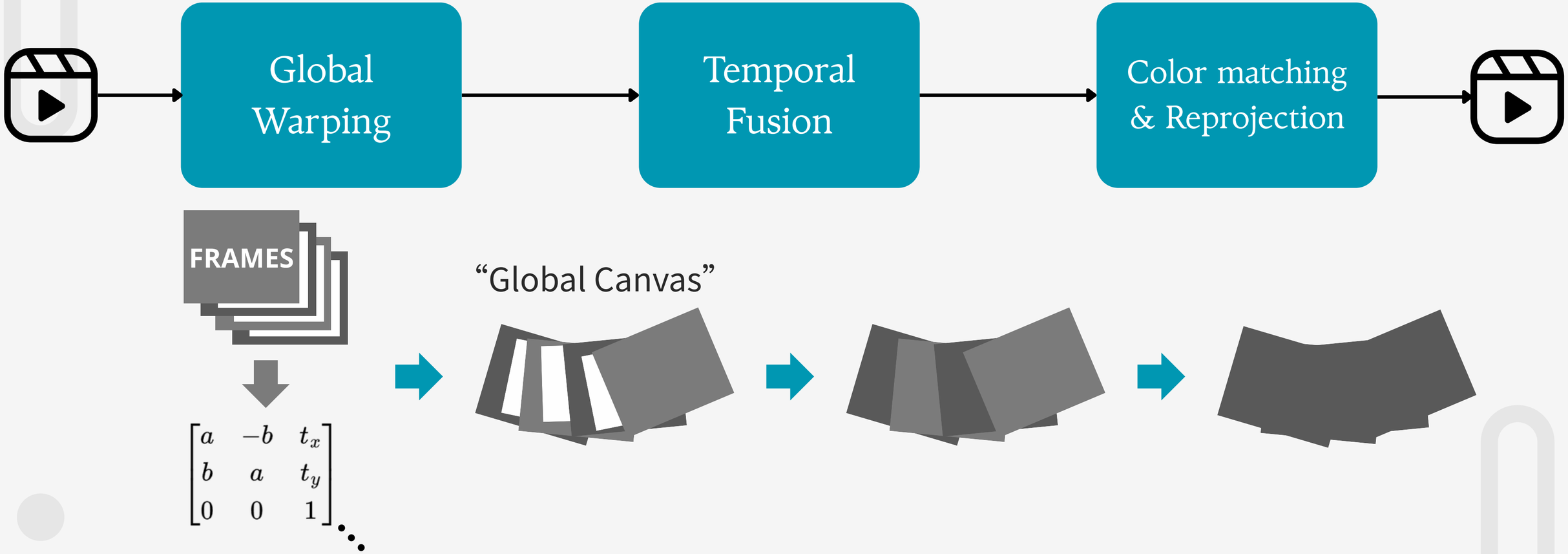
Workflow

01 Preliminaries

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3-Stage Pipeline



1. Global Warping

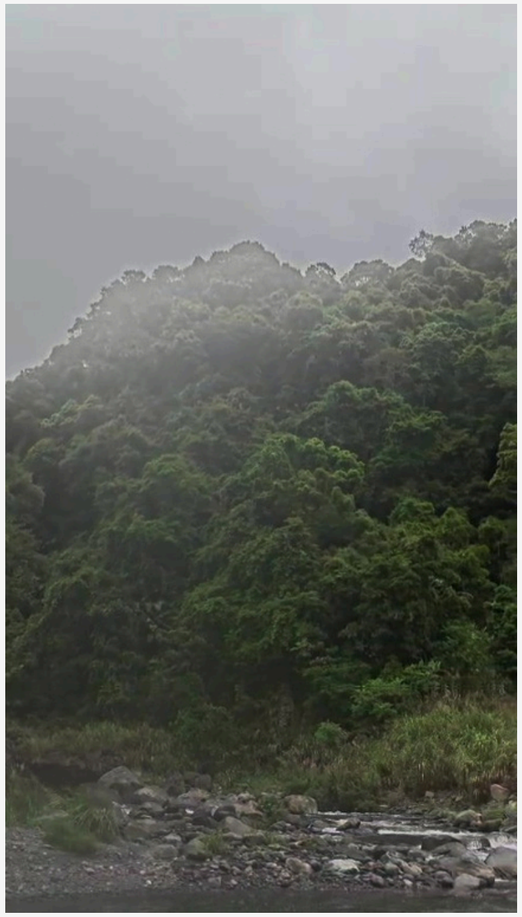
I. Naive Focus Mask

a. Ignore the top and bottom 25% of the frame

i. top areas: sky and moving clouds

ii. bottom areas: water flow

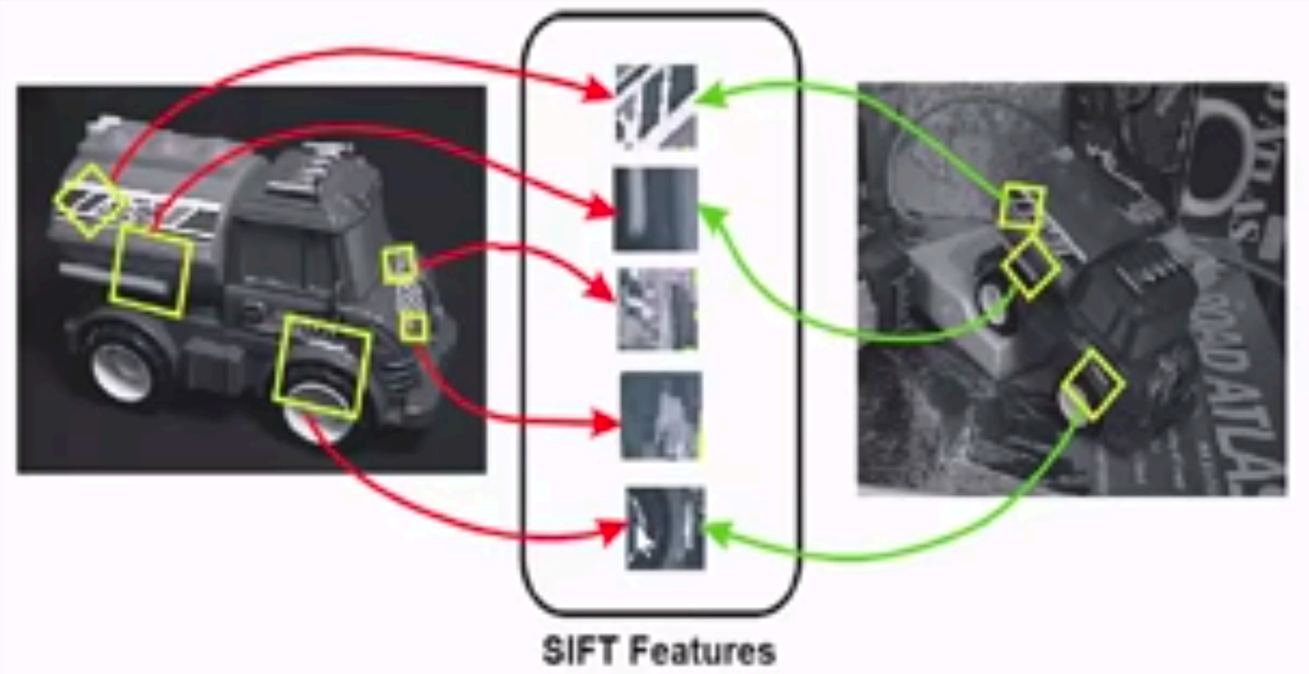
b. Focus on central rigid structures



1. Global Warping

II. SIFT(Scale-Invariant Feature Transform)

a. Extract up to 2,000 keypoints per frame



1. Global Warping

III. FLANN: Fast Approximate K-NN

a. Match keypoints between the Current Frame and the Target Frame

b. each feature descriptor is matched to n number of closest neighbors.

(n=2)

c. faster computation: O(nlogn) (v.s. BruteForceMatcher, O(n^2))

1. Global Warping

IV. Affine Transformation
    a. Calculate the geometric displacement between frames
    b. 4-DOF
        i. Restrict transformations strictly to Translation (X, Y), Rotation, and Scaling
        ii. Shape Preserving: Maintain geometric integrity to ensure sharp edges during the Median fusion stage
    c. Error Mitigation
        i. Hard-clamp the calculated scale ratio within [0.98, 1.02]
        ii. Avoid abnormal "breathing effects" or sudden zooming caused by feature mismatches





1. Global Warping

V. Global Alignment Strategy

a. Keyframe Anchoring

i. Compare the current frame against a fixed keyframe anchor rather than just the previous frame

ii. When accumulated translation is too large or good matches are too few → Reset the keyframe

b. Middle Plane Transfer

i. Shift the global coordinate origin to the middle frame to evenly distribute projection distortion









1. Global Warping

VI. Gaussian Trajectory Smoothing

a. Eliminate the jello effect
b. Deconstruct all transformation matrices H into physical parameters
    i. X Translation, Y Translation, Rotation Angle, and Scale Ratio
c. Apply a 1D Gaussian Filter to each parameter independently, then reconstruct the final smoothed H matrices

smoothed



2. Temporal Fusion

Initialize global canvas

Project every frames onto global canvas

Pixelwise filtered-median operation



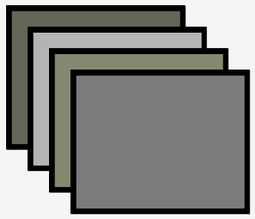


Project

FRAMES

$$\begin{bmatrix} a & -b & t_x \\ b & a & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

For any pixel:



Get %

top 5% unlikely foggy

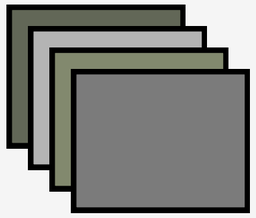
Fog Confidence = a

top 5% likely foggy

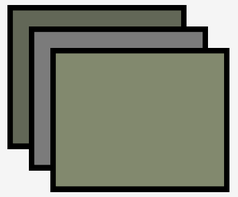
Fog Confidence = b

Fog Confidence Threshold = a+(b-a)*0.25

Whiteness Threshold = 220



Filter



median

final pixel value

Pixelwise



2. Temporal Fusion

Calculation of Fog Confidence consists of three component:

- Dark Channel Prior score: according to paper Single Image Haze Removal Using Dark Channel Prior, pixels with less Dark Channel Prior value are less likely to be affected by fog.

$$conf_{dcp}(x) = \min_{y \in \Omega(x)} \left(\min_{c \in \{R,G,B\}} I^c(y) \right)$$

- Gradient magnitude score: pixels affected by fog often look smoother, that is, have less gradient magnitude.

$$G(x) = \sqrt{(\nabla_X I_{gray}(x))^2 + (\nabla_Y I_{gray}(x))^2} \quad conf_{grad}(x) = 1 - \min\left(1, \frac{G(x)}{P_{95}(G) + \epsilon}\right)$$

- Saturation score: pixels affected by fog often have less saturation.

$$conf_{sat}(x) = 1 - S(x)$$

Then we calculate Fog Confidence Score by weighted averaging three of them

3. Color matching & reprojection

- Reinhard coloring matching (in Lab space)
- EMA (Exponential Moving Average), α = 0.1

$$\mu_t \equiv \alpha\mu_{t-1} + (1-\alpha)\mu_{\text{raw}} \qquad \mu_{\text{bg}} \mapsto \mu_t$$
$$\sigma_t \equiv \alpha\sigma_{t-1} + (1-\alpha)\sigma_{\text{raw}} \qquad \sigma_{\text{bg}} \mapsto \sigma_t$$

Demo

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